Exercise 7

On absolute maxima and minima (Absolute extrema)

From the school book | Understand | Apply | Higher Order Thinking Skills

1 **Find the extrema values of the following function on the given interval :**

(1) $f(x) = 3 - 5x - x^2$ $[-4, 3]$

(2) $y = x^3 - 9$ $[0, 2]$

(3) $f(x) = x^4 + 2$ $[-1, 2]$

(4) *(1st Session 2018)* $f(x) = x(x^2 - 12)$ $[-1, 4]$

(5) *(2nd Session 2008)* $f(x) = x^3 - 3x^2$ $[1, 4]$

(6) *(2nd Session 2005)* $f(x) = x^3 - 9x^2 + 24x - 1$ $[0, 5]$

(7) *(2nd Session 2000)* $f(x) = x^3 - 9x^2 + 15x$ $[-2, 4]$

(8) *(1st Session 2003)* $f(x) = x^3 + x^2 - 5x - 5$ $[-\sqrt{5}, 0]$

(9) $f(x) = (x - 1)(x - 2)^2$ $[0, 5]$

(10) $f(x) = 2x - 3x^{\frac{2}{3}}$ $[-1, 3]$

(11) *(1st Session 2017)* $f(x) = 3x^4 - 4x^3$ $[-1, 2]$

(12) $f(x) = 2x^4 - x^2 + 1$ $[-1, 1]$

2 **Choose the correct answer from those given :**

(1) If the function f is defined on the interval $[a, b]$ and if $f'(x) < 0$ on the same interval , then the absolute maximum value of $f(x)$ in this interval = ...

(a) $f(a)$ (b) $f(b)$ (c) a (d) b

(2) If the function f is defined on the interval $[a, b]$ and if $f'(x) > 0$ on the same interval , then the absolute minimum value of $f(x)$ in this interval = ...

(a) $f(a)$ (b) $f(b)$ (c) a (d) b

(3) *(Speciman 2025)* The absolute maximum value of the function :

$f : f(x) = x^2(x + 3)$ on $[-1, 1]$ is .....

(a) 4 (b) 3 (c) 2 (d) 0

(4) *(2nd Session 2019)* If the function $f : f(x) = 2a x^2 + b x + 3$ has a local extrema at $(1 , 2)$, then $a + b =$ .....

(a) -1 (b) $\frac{5}{2}$ (c) $-\frac{3}{2}$ (d) $\frac{3}{2}$

(5) If $x \in [0 , \pi]$, the function $f : f(x) = x \sin x + \cos x$ has absolute minimum value at $x =$ .....

(a) zero (b) 1 (c) π (d) $\frac{1}{2}\pi$

(6) The maximum value of the function $f : f(x) = 3 - \sin x$ is ..... where $x \in \mathbb{R}$

(a) 1 (b) 2 (c) 3 (d) 4

(7) *(2nd Session 2022)* The absolute minimum value of the function $f : f(x) = -\frac{1}{3}x^3 - x^2 + 9$, where $x \in [-3 , 3]$ equals .....

(a) -9 (b) $-\frac{23}{3}$ (c) $\frac{23}{3}$ (d) 9

(8) *(1st Session 2022)* The absolute maximum value of the function $f : f(x) = \frac{x^2}{3 - x} + 1$, $x \in [-1 , 2]$ equals .....

(a) $\frac{5}{4}$ (b) $\frac{4}{3}$ (c) 5 (d) 6

(9) If $f : \mathbb{R} \longrightarrow \mathbb{R}$ where $f(x) = x^3 + 3x^2 - 9x$ and a , b are the absolute minimum and maximum values of function f respectively on the interval $[-2 , 2]$, then $b - a =$ .....

(a) 17 (b) -17 (c) 27 (d) -27

(10) If $f : f(x) = \cos x$ where $x \in [\frac{-\pi}{2} , \frac{5\pi}{2}]$ has an absolute maximum value , then the number of times it reaches to the maximum values is .....

(a) 1 time. (b) 2 times. (c) 3 times. (d) 4 times.

(11) *(Speciman 2025)* If the function $f : [-3 , -1] \longrightarrow \mathbb{R}$ where $f(x) = x + \frac{a}{x}$ is increasing on $]-3 , -1[$ and the absolute maximum value of f equals -2 , then $a =$ .....

(a) -2 (b) -1 (c) 1 (d) 2

(12) *(1st Session 2025)* The absolute maximum value of the function $f : f(x) = x + \frac{4}{x}$ on the interval $[1 , 3]$ equals .....

(a) -4 (b) 4 (c) $\frac{13}{3}$ (d) 5

(13) *(2nd Session 2021)* The absolute maximum value of the function $f(x) = x \ln x$ where $x \in [e^{-2} , e]$ equals .....

(a) $-2e^{-2}$ (b) e (c) 2 e (d) $-e^{-1}$

(14) *(Speciman 2025)* The absolute maximum value of the function $f : f(x) = x\,e^{-x}$ in the interval $[0 , 2]$ equals .....

(a) 0 (b) $\frac{1}{e}$ (c) $\frac{2}{e}$ (d) 1

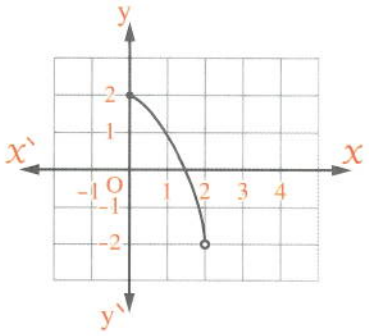
(15) The function in the opposite figure ..... where $x \in [0 , 2[$

(a) has absolute minimum value and absolute maximum value.

(b) has absolute minimum value but not absolute maximum value.

(c) has absolute maximum value but not absolute minimum value.

(d) has neither absolute maximum value nor absolute minimum value.



3 If $f'(x) = a\,x + b$ where $a < 0$ investigate the existence of a extrema value for the function f and show its type if found.

4 *(1st Session 2023)* Find the absolute minimum value of the function $f : f(x) = \sqrt[3]{(x-8)^2} + 1$, where $x \in [0 , 9]$ « 1 »

5 **Find the absolute maximum and minimum value of each of the following functions on the given interval :**

(1) *(1st Session 2012)* $f(x) = \frac{x^2 + 9}{x}$ $[1 , 6]$

(2) 🕮 *(2nd Session 2018)* $f(x) = \frac{4x}{x^2 + 1}$ $[-1 , 3]$

(3) *(1st Session 2004)* $f(x) = x + \frac{1}{x}$ $\left[\frac{1}{2} , 3\right]$

(4) $f(x) = \frac{x^2 + x + 1}{x^2 - x + 1}$ $[-2 , 2]$

6 *(Trial 2023)* If $f(x) = a\,x - x^3$ where $x \in [0 , 4]$ and $f(1)$ is the absolute maximum value of the function , then find its absolute minimum value of the function where a is constant.

7 **Find the extrema values of the following functions on the given interval :**

(1) $f(x) = \sin x$ $\left[\frac{5\pi}{6} , \frac{11\pi}{6}\right]$

(2) 🕮 $f(x) = \sin x + \cos x$ $[0 , 2\pi]$

(3) $f(x) = 2\sin x + \cos 2x$ $\left[0 , \frac{\pi}{2}\right]$

(4) *(2nd Session 2017)* $f(x) = 10\,x\,e^{-x}$ $[0 , 4]$

(5) $f(x) = x\,e^{x}$ $[-3 , 1]$

(6) $f(x) = x^2\,e^{-x}$ $[-1 , 3]$

8 **Find the extrema values of the following functions on the given interval :**

(1) $f(x) = \sqrt{x-1}$ $[2 , 5]$

(2) $f(x) = \sqrt[3]{x}$ $[-8 , 8]$

(3) $f(x) = \sqrt{16 - x^2}$ $[-4 , 3]$

(4) $f(x) = \sqrt[3]{(x-1)^2}$ $[0 , 9]$

(5) $f(x) = x\sqrt{1-x}$ $[-1 , 1]$

(6) $f(x) = x - \sqrt{x}$ $[0 , 9]$

(7) $f(x) = \sqrt{8x - x^2}$ $[0 , 8]$

(8) $f(x) = \sqrt[3]{x^2 - 4}$ $[-1 , 4]$

(9) $f(x) = \dfrac{x}{\sqrt{x-1}}$ $\left[\frac{5}{4} , 5\right]$

(10) $f(x) = \dfrac{\sqrt[3]{x-2}}{x}$ $[1 , 4]$

9 **Find the extrema values of the following functions on the given interval :**

(1) $f(x) = \begin{cases} 2x + 3 & , \ x < 1 \\ x^2 + 4 & , \ x \geq 1 \end{cases}$ $[-2 , 2]$

(2) $f(x) = \begin{cases} (x+3)^2 & , \ x < 0 \\ (x-3)^2 & , \ x \geq 0 \end{cases}$ $[-2 , 2]$

(3) $f(x) = \begin{cases} x^2 + x - 2 & , \ x \leq 2 \\ 5x - 6 & , \ x > 2 \end{cases}$ $[-1 , 3]$

(4) *(2nd Session 2002)* $f(x)=\begin{cases}5-x^2 & , \ x\le 2\\ x^2-3 & , \ x>2\end{cases}$ $[-3\ ,3]$

(5) 📖 $f(x)=\begin{cases}x^3-3x^2 & , \ x\le 0\\ x^2-2x & , \ x>0\end{cases}$ $[-3\ ,3]$

(6) *(2nd Session 2012)* $f(x)=\begin{cases}3x^2+12x-1 & , \ -1\le x\le 2\\ 37-x & , \ 2<x\le 3\end{cases}$ $[-1\ ,3]$

(7) $f(x)=|x|+2$ $[-1\ ,2]$

(8) $f(x)=3-|x-1|$ $[0\ ,3]$

(9) *(1st Session 2009)* $f(x)=(x-4)|x|$ $[1\ ,4]$

(10) *(2nd Session 2013)* $f(x)=x|x-2|$ $\left[\frac{1}{2}\ ,\frac{5}{2}\right]$